Final lab test

AI

1. **Problem Statement:** You are given a graph where nodes represent cities and edges represent roads between them. Each road has a cost (like distance or travel time). Your task is to find the shortest path from a starting city to a destination city using both DFS and Best-First Search (BFS).

Tasks:

1. **Graph Representation:**
   - Choose a data structure (e.g., adjacency list) to represent the graph.
   - Create a graph with at least 10 cities and the corresponding roads with costs.
2. **Algorithm Implementation:**
   - Implement the DFS algorithm to find a path from the starting city to the destination city.
   - Implement the Best-First Search algorithm using a heuristic based on distance to prioritize paths.
3. **Pathfinding:**
   - For both algorithms, output the found path and the total cost.
   - If no path exists, indicate this clearly.
4. **Performance Analysis:**
   - Compare the two algorithms based on:
     - Path cost (length)
     - Number of nodes explored
     - Memory usage
5. **Discussion:**
   - Discuss the strengths and weaknesses of DFS and Best-First Search.
   - In which situations might you choose one algorithm over the other?

Submission Requirements:

- A report detailing your methods, findings, and performance analysis.
- Code for both DFS and Best-First Search implementations.
- Optional: Visuals or diagrams of the graph and paths.

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